Dong Quan Tran

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EDUCATION

University of Texas at Arlington (UTA) Expected: December 2027

B.S. in Computer Science, Minor in Mathematics *GPA: 4.0 / 4.0*

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases, Operating Systems, Data Mining, Computer Networks, Cloud Computing, Information Security, Artificial Intelligence

TECHNICAL SKILLS

Programming: Python, C++, C, Git, Linux, SQL, TypeScript, React
Frameworks: PyTorch, FastAPI, Pandas, TensorFlow, Ollama, Meta Muse API, Gemini API, scikit-learn, Three.js
Cloud & Infrastructure: Amazon Web Services (AWS), Vultr, Docker, REST APIs, Websockets (ws)
Concepts: Agile Methodologies, Cybersecurity, Data Analysis (Python), Machine Learning, Computer Vision, Network Security, Anomaly Detection, Threat Analysis, Website Fingerprinting

WORK EXPERIENCE

Cybersecurity Research Assistant | *Information Security Lab, UTA* Arlington, TX | Sep. 2025 – Present

- Research **threat analysis - Website Fingerprinting** on **Tor Browser** across **100+** experiments under **Dr. Remi Chou** to evaluate user deanonymization risks and privacy–performance tradeoffs.
- Build **Python/Linux pipelines** to automate traffic collection across **100+** websites, gathering **100,000+** traffic traces, reducing manual experiment configuration by up to **40%**.
- Lead a **five-person** research team as an **ACM Research Officer**, directing experimental design, execution, and analysis; coordinate weekly faculty-advisor reviews for evaluation and progress updates.
- Presented findings at **2026 UTA Innovation Day (170+ projects, 400+ attendees)**, earning an **extension offer** to continue research in the **UTA REU program**.

CX – ITS Technical Assistant | *Office of Information Technology, UTA* Arlington, TX | Dec. 2025 – Present

- Provide front-line **technical support** across **6 c**ampus computer labs, troubleshooting account access, software, networking, and printing; contribute to a **30% increase** in positive support feedback.
- Maintain **300+ lab workstations** through **bi-weekly** software updates, deployment, reimaging, and system reboots; personally resolved **400+ support tickets**.

Instructional Math & Physics Tutor | *Math Resource Center, Tarrant County College* Hurst, TX | Aug. 2024 – May 2025

- Delivered **500+** hours of **one-on-one** and **group** Engineering Math and Physics instruction to **200+** students, with **95%** reporting improved comprehension and academic performance.
- Led **11** students in the district-wide **TCC Jim Bolen Math Competition (200+ participants)**, personally earning **1st Place 2 years in a row** with **8+** scholarships and awards, growing student participation by **over 30%.**

PROJECTS

Let Him Rook – Real-Time Online PvP 3D Chess | *TypeScript, Three.js, MediaPipe, Meta Muse Spark* Oct. 2026 - Present

- Led a **four-person** team at **HackTX 2026** to build a **real-time multiplayer 3D chess game** with **6 webcam-controlled** combat classes, server-authoritative WebSocket networking, and a **60 Hz** combat simulation.
- Optimized **MediaPipe** tracking and rendering across **Windows** and **macOS** using **Web Workers**, **GPU inference**, and **Apple Silicon CoreML** acceleration; increased tracking **FPS** from **8** to **60 (7.5×).**
- Deployed a **Node.js** multiplayer server on **Vultr**; integrated **Meta's Muse Spark API** for **contextual gameplay advice** with timeouts and fallback responses to prevent AI failures from interrupting gameplay.

InferSQL – Analytical SQL Engine with Text-to-SQL Copilot | *Ollama, Gemini API, DataFusion, SQLGlot* Apr. 2026 – Jul. 2026

- Built an **Apache Arrow/DataFusion analytical SQL platform** with modular query planning and schema validation, supporting joins, nested subqueries, aggregations, and set operations across **200+** tests.
- Integrated a guarded **Google Gemini/Ollama text-to-SQL copilot** with schema grounding, read-only validation and repair retries; passed **33/33** curated evaluations across **10** categories.
- Achieved **6.7 ms** filters**, 8.9 ms** sorts**, 45.3 ms** aggregates**, 44.6 ms** joins across **1,000,000-row** benchmark workloads.

SentinelTI – Threat-Intelligence URL Classifier | *Python, FastAPI, React, XGBoost, SQLite, URLhaus* Feb. 2026 – Present

- Built an **end-to-end ML and heuristic URL threat-scoring platform**, classifying benign, suspicious, and malicious URLs at **1,110** URLs/sec with **1.0** ms p95 core-scoring latency.
- Integrated **URLhaus** intelligence into **SQLite** and built **700+ Pytest** cases, validating API authentication, rate limiting, model loading, scoring, and dataset integrity, supporting **regression-safe** refactors.
- Benchmarked **Logistic Regression, XGBoost**, and **LightGBM** on **450,000+** URLs; XGBoost achieved **0.999** ROC-AUC, **0.999** average precision, and **99.08%** malicious recall on a **135,000-URL** test set**.**